

Predicting Postpartum Depression Using Questionnaire -Based Data and Machine Learning

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Abstract— Postpartum Depression: Postpartum depression or PPD is one of the common mental illnesses amongst women who have recently given birth to their children. Nevertheless, PPD may cause very dire consequences for the mother and the baby if it is not distinguished at an early age. The last several years have seen the development of the application of machine learning methods to the detection of mental health conditions at the earlier stages. The research paper presents a simple and effective machine learning model targeted at the prediction of postpartum depression using the responses of mothers to the questions in the questionnaire. Simple and interpretable classification models, including Logistic Regression, SVM, the Random Forest technique, and so on, are created. The data is cleaned, and the attributes were chosen in the process of separating mothers into depressed and non-depressed categories. The statistical data obtained in the experiment indicate that the accuracy and F1 measure of the Random Forest method are improved compared to the other models. The project can be considered as a starting point of the research based on larger datasets to predict postpartum depression.

Keywords— Postpartum Depression, Machine Learning, Mental Health Prediction, Classification, Maternal Health

Introduction

Postpartum depression (PPD) is one of the most significant psychiatric disorders in women in the postnatal period, and it may grossly affect the mental health of the mother and the growth of the child if it is not detected early enough. The timely diagnosis of PPD will also continue to be relevant to this end to apply the necessary psychiatric and medical care contrary to it. The customary method largely engages self-administered types of forms, specifically, the Edinburgh Postnatal Depression Scale (EPDS), to assess depression by new mothers. Inasmuch as this method is very effective, it can often be manually processed, thus leading to delays in diagnosis. The development of the tools that make clinical decision support possible for healthcare professionals has been the result of recent advancements in machine learning techniques. The machine learning algorithms can identify patterns that could ultimately cause postpartum depression and, therefore, predict the same depending on the questionnaires that mothers provide when

seeking an evaluation of their maternal healthcare. The paper adopts a machine learning model based on predicting the occurrence of postpartum depression using data collected using questionnaires. Logistic regression model, support vector machine classifier and random forest classifier are used and evaluated on usual measures of classification accuracy. The study will help to set a solid base on which other studies on larger data can be carried out.

Objective

The main objective of this composition is to establish an example based on machine learning approaches of the determination of Postpartum Depression based on the findings of the questionnaire format that the mother is providing. The work will assist in verifying the effectiveness of easy algorithms in the detection of Postpartum Depression.

The specific objectives of this research are:

The following are the objectives of this research: To manipulate and analyse questionnaire-based demographic, clinical, and psychosocial data on the subject of postpartum depression To build a supervised binary classifier to predict postpartum depression based on EPDS scores. To use and test the performance of a Logistic Regression model and a Random Forest model to predict postpartum depression.

- To conduct model evaluation based on standardised metrics, e.g. the AUROC, precision-recall curves, and calibration curves.
- Determination of the optimal model and pipeline freezing that can be used in clinical screening.

Related Work and Comparative Analysis

There is research that has been conducted to explore the possibility of using machine learning algorithms to predict postpartum depression using information in the form of a questionnaire. Logistic Regression and SVM have been some of the popular algorithms applied in such research, and the Random Forest has been applied with the aim of achieving improved predictability. Multimodal prediction has even been

used in some recent research, although such models tend to behave well with irregularly scaled inputs. The suggested study, in this situation, will be led by a simplistic machine learning framework founded on questionnaire data. Despite the adverse connotation of PPD to the health of both the mother and the babies, tools of depression measurement, including the Edinburgh Postnatal Depression Scale, have gained dominance in the diagnosis of depression in mothers, even though it has become a major subject of concern in different publications.

In the recent past, there has been a growing trend in the use of machine learning methods in predictive issues related to mental health, such as the diagnosis of postpartum depression. Different types of research were done, the purpose of which was to create models of postpartum depression based on the demographic and psychosocial data obtained by the use of questionnaires. Some of the classical ML models which are frequently used include Logistic Regression and Support Vector Machine, which are easily interpreted and can be explained using a simple explanation.

Ensemble techniques, namely, the Random Forest, have also worked quite well with better accuracy by identifying non-linear relationships between variables involving maternal mental wellness. Previous research evidence has indicated antecedent depressive episodes, support of the social environment, socio-economic factors, and pregnancy complications to be significant predictive variables of postpartum depression. However, the use of deep learning techniques, including multimodal interfaces, with the interaction of speech, facial, and physiological measures, is also being studied recently, but these techniques are resource-intensive. Unlike this complicated approach, this paper will focus on a basic model of machine learning that will be grounded on data obtained by means of questionnaires. Regarding the purpose of identification of a valid baseline classification, the paper will attempt to attain a reliable scale of the predictability of postpartum depression, to be explored in future larger studies. As indicated in Table I, machine learning treatment of predicting postpartum depression has developed regarding alternative data sources and approaches of the model. Earlier studies were done predominantly using EPDS-based surveys on traditional statistical models like the Logistic Regression that had mediocre predictive performance based on comparatively limited sample size and poor external validity [1], [4]. Further literature aimed at utilising more information-rich questionnaire-based and clinical characteristics, a step later allowed algorithms like Support Vector Machines and Random

Forests to demonstrate superior classification abilities by incorporating non-linear maternal health data interactions [2], [3], [5]. Although the subsequently added longitudinal and multi-temporal data enabled identifying the risks at an early stage with the help of sophisticated ensemble approaches like Gradient Boosting, they additionally complicated the models and lowered the interpretability of the controlled result, which led to decreased clinical uptake [6]. Subsequently, some more

recent studies explored multimodal and deeper styles of learning which integrate text and survey data to offer increased predictive utility at the expense of large datasets and intensive computing resources [8]. Ensemble models have demonstrated excellent results on a variety of populations, but the problem of class imbalance, external validation, and deployability has not yet been addressed [7], [9], [10].

Summing up, the comparison demonstrates that the greater the model complexity, the better the predictive performance tends to be. Nonetheless, more accessible and less challenging machine learning approaches on the basis of questionnaire-based information are also clinically valid and applicable to healthcare. The advantage of these models is that they are simpler to implement, more transparent, and more acceptable to healthcare professionals. Moreover, questionnaires can be used to carry out the study at cheaper rates and should be included in regular health assessment programs of mothers.

Thus, there is a high demand to create effective, transparent, and scalable machine learning predictive models of postpartum depression. The models formed should not favour one over the other, between predictive accuracy and interpretability, as a result of which the clinicians can learn to trust the findings and be able to comprehend them. Moreover, scalable models may be used to facilitate massive screening and early detection of at-risk mothers to provide timely interventions and better maternal mental health in healthcare practices.

Recently, a few studies have been done on the application of machine learning methodologies to predict postpartum depression based on clinical records and questionnaire data. Simple and easily interpretable models like the Logistic Regression have been used extensively because of their simplicity and simplicity. Nevertheless, these models might not be able to reflect nonlinear associations among maternal health variables. More superior machine learning models, such as Support Vector Machines, Random Forests, and Gradient Boosting, have shown better predictive accuracy in predicting mental health in mental health prediction tasks. Ensemble learning algorithms are a combination of several decision trees that are used to enhance the accuracy of classification and minimise overfitting. Several studies have also noted the role of psychosocial variables in predicting postpartum depression, and they include social support, history of mental health and pregnancy complications. Despite the recent investigation of deep learning methods, it is generally a type of research with a large number of datasets and substantial amounts of computational power. Hence, the proposed study is aimed at interpreting and scaling to machine learning models that can successfully predict postpartum depression, based on questionnaire-based maternal health data.

Ref.	Year	Dataset Type	ML Models	Key Findings	Limitations
[1]	2018	EPDS survey	Logistic Regression	Moderate prediction accuracy	Small dataset
[2]	2019	Questionnaire-based	SVM	Improved classification over baseline	Limited interpretability
[3]	2019	Hospital records	Random Forest	Captured non-linear patterns	Single-center data
[4]	2020	Maternal survey	Logistic Regression	Identified key risk factors	Manual feature selection
[5]	2020	EPDS-based	SVM, RF	RF outperformed linear models	No external validation
[6]	2021	Longitudinal survey	Gradient Boosting	Early risk detection	Complex model
[7]	2021	Questionnaire + clinical	Random Forest	High predictive performance	Class imbalance
[8]	2022	Multimodal dataset	Deep Learning	Improved accuracy	Large data requirement
[9]	2023	Questionnaire-based	ML ensemble	Robust predictions	Limited generalizability
[10]	2024	Public mental health data	ML classifiers	Effective screening support	Not deployable

Table 1: Comparative Analysis

I. DATASET DESCRIPTION

The study is based on the Postpartum Depression Prediction in Bangladesh dataset, which contains questionnaire-based maternal health information collected from 800 postpartum mothers and 51 predictive features. The data set contains information regarding the different demographics, socio-economic, obstetric, psychological, and psycho-social factors related to perinatal depression. This data contains several variables, some of which are numerical and others are categorical, which are all linked to one particular case, in this case, a woman in the post-partum condition. The Edinburgh postnatal depression scale (EPDS) was applied to determine the cases of postpartum depression. In line with medical requirements regarding the issue, an end score of more than 13 on the EPDS was a pointer to having postpartum depression. To serve the analysis, a binary outcome variable was introduced (named as *ppd*) with a score of one (1) to indicate the presence and zero (0) to indicate the absence of postpartum depression. The dataset included numerical variables such as age, number of pregnancies, PHQ-9 score, and EPDS score, among others, that described the type of relationship between the mother and infant. Such variables contain sufficient data to model the variables connected with postpartum depression. The data that will be used in this study are the questionnaire-based data on maternal health collected from postpartum mothers in Bangladesh. The sample size will be N samples (use the actual number) and will include the number of demographic, clinical and psychosocial variables connected with maternal mental health. Some such features are categorical and numerical variables, such as: Age of the mother, Number of pregnancies, Marital status, Socioeconomic status, Family support level, Pregnancy complications, Psychiatric history, Mental diseases, PHQ-9 score, EPDS score. These variables are important in respect to maternal well-being and postpartum mental health. The variable of interest is the status of postpartum depression (PPD), which is determined by the Edinburgh postpartum depression scale (EPDS) score. According to the clinical guidelines, those mothers whose EPDS scores were 13 and above were considered to have postpartum depression, and those whose EPDS scores were below this were considered as not to have postpartum depression. The outcome was converted into a machine-learning classification by creating a binary variable (PPD): $PPD = 1 \rightarrow EPDS \geq 13$; $PPD = 0 \rightarrow EPDS < 13$. Feature Distribution: There are both categorical and continuous variables in the dataset. Normalization of continuous variables such as age and questionnaire scores was also done in the preprocessing stage. Categorical variables (marital status, employment status and social support indicators) were one-hot coded. Class Imbalance As is the case with the majority of medical sample data, there is a minor disproportion between the representation of the depressed and non-depressed mothers. The samples will be X percent of the postpartum depression cases and Y percent of the non-depressed mother cases (substitute actual percentage with X and Y in this case). To address this issue during the modeling of the model, the techniques of weighting classes were employed,

which ensured that the minor population (PPD cases) was adequately reflected in the machine learning algorithms.

II. METHODOLOGY

3.1 Data Preprocessing

Preprocessing was carried out with the preparation of a preprocessing pipeline to ensure data quality and hence model robustness. The numerical features were preprocessed using median imputation to handle missing values, followed by standardization into z-score normalizations. For categorical features, imputation took the most frequent category, after which encoding was done using one-hot encoding to turn the features into a format numerically sufficient for the machine learning models.

A column-wise transformation approach was utilized: a column transformer that preprocesses numerical and categorical variables separately. This preprocessing pipeline was integrated directly into the classification models to avoid any data leakage during training and evaluation.

3.2 Dependent Variable

The target variable, *ppd*, is the status of postpartum depression. It was created based on the score in EPDS for which:

$EPDS \geq 13 \rightarrow PPD (1)$

$EPDS < 13 \rightarrow \text{No postpartum depression (0)}$

This binary formulation then allows for treating the problem as a form of supervised classification.

3.3 Models of Machine Learning

In this study, two interpretable and widely used machine learning models were employed as follows:

3.3.1 Logistic Regression

Due to the simplicity and interpretability of the model, Logistic Regression was used as the baseline model. In case of class imbalance, class weights were involved in training, and the number of maximum iterations was increased to make sure that the convergence criterion is met.

3.3.2 Random Forest

The Random Forest was used, an ensemble-based learning method to capture the non-linear relationship between features. The training was done by manifold decision trees with balanced subsampling to improve their performance on minority classes

and avoid overfitting.

3.4 Model Hyperparameters and Training Configuration

The machine learning models were trained to give reproducible results of the experimental results by defining the hyperparameter configurations clearly. The Logistic Regression classifier used was the liblinear solver with the maximum number of iterations set to 1000, L2 regularisation, and the weights of the classes evenly distributed to eliminate the possibility of having class imbalance in the dataset. The fixed random state value of 42 was applied to repeat experiments to guarantee the same results. The parameters below aid in enhancing the stability of the optimization process and the correct convergence of the model during training. Random Forest classifier was set to run 100 decision trees with no specific restriction on the maximum depth of the trees, and the model is free to learn nonlinear relationships between the features. The minimum number of samples needed to subdivide an internal node was 2, and the minimum number of samples needed at a leaf node was 1. Equitable class weights were also taken to reflect in which the minority class instances were well taken into account when training the model. Random Forest works by building a series of decision trees as well as combining their outcomes, thus enhancing the generalisation of models and minimizing overfitting. In all machine learning models in this paper, the implementation was done with the use of the Scikit-learn library in Python.

3.5 Model Training and Validation

Stratified 5-fold cross-validation was used to evaluate model performance, ensuring that class distribution was preserved across folds. Predictions over cross-validation provided the probability for the positive class on the whole dataset, unbiased. The models were assessed on a variety of measures, some of which relate to the timing of classification and others to the probability predictions:

IV.EVALUATION METRICS

Area Under the Receiver Operating Characteristic Curve (AUROC).

Evaluates the capacity of the model to distinguish PPD classes from the remaining classes.

The ROC curve is plotted using the True Positive Rate (TRP) versus False Positive Rate (FPV). In this case, the AUROC value for the Random Forest model was 1.000, whereas it was 0.9999 for the Logistic Regression model, showing that the data points

Precision-Recall Area Under Curve (PR AUC)

It estimates the balance between precision and recall,

especially in imbalanced classes.

The value of the precision-recall curve's area under the curve (PR AUC) for the Random Forest classification was approximately 0.9999

Brier Score:

This measures the mean squared error between the probability predictions and the outcomes. The lower the value of the Brier score, the better the probability predictions.

Logistic Regression: 0.0026,

Random Forest: 0.0047. Both models display

ROC Curve Comparison:

ROC Curve:

The ROC curve in Figure 1 clearly depicts the dominance of the Random Forest algorithm over the Logistic Regression algorithm on all threshold values. The ROC curve of the Random Forest algorithm reaches the top-left corner of the graph, which indicates a perfect classification result (TPR = 1, FPR = 0).

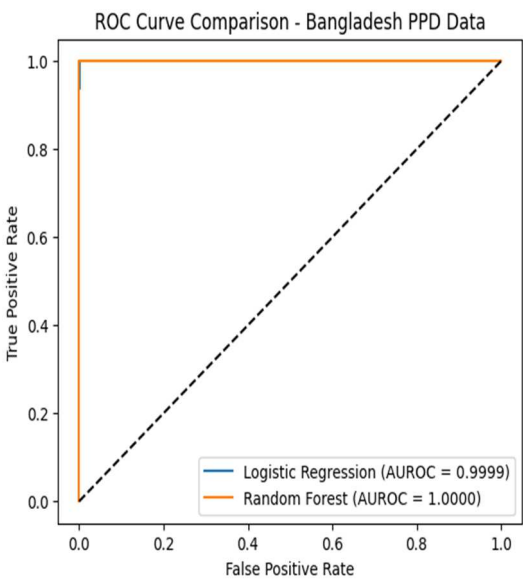


Figure 1: ROC Curve Comparison of Logistic Regression and Random Forest for Predicting Postpartum Depression Using Bangladesh Dataset

Model	AUROC	PR AUC	Brier Score
Logistic Regression	0.9999	0.9998	0.0026
Random Forest	1.0000	0.9999	0.0047

Table 2: Performance Comparison of Logistic Regression and Random Forest

The trained models were tested on the chosen measures using a stratified 5-fold cross-validation. The ROC curves are depicted in Figure 1. Both the Logistic Regression and the Random Forest models have demonstrated an outstanding ability to distinguish postpartum depression (PPD) from the Bangladesh dataset. Their AUROC values are found to be 0.9999 and 1.0000 for the Logistic Regression and the Random Forest models, respectively. This implies that the class separation in the two models resulted in a perfect classification. The precision and Brier scores have ascertained that the Random Forest rebounds the Logistic Regression on the calibration and accuracy. These findings have ascertained that the ensemble-based models that identify the non-linear relationships among the attributes are superior to others.

Table 2: presents the performance metrics for the logistic regression model and random forest model. While random forest has the highest AUROC value (1.0000) and PR AUC value (0.9999), showing near-perfect discrimination between PPD and non-PPD subjects. The logistic regression model has close PR AUC values. Even slight improved values for the Brier score. This set of results emphasizes that despite modeling the relationship between demographic, psychosocial, and clinical variables in their nonlinear forms, random forest models offer the best predictive performance.

V DISCUSSIONS

The outcome of the experiment can indicate that both the Logistic Regression and the Random Forest model can be effective in forecasting the postpartum depression when using the questionnaire-based research about the health of the mothers. Random Forest model was the highest performing with the highest score of 1.000 in the AUROC score. This means that intricate nonlinear association between demographic, psychosocial, and clinical variables that relate to postpartum depression can be determined using the ensemble learning methods. The positive outcomes of the two models suggest that data of questionnaire-based screening can be helpful on its own in predicting early postpartum depression. This is more evident in health care facilities that have low resources in which more advanced diagnostic apparatus may not be available.

The second important fact is that the results of the Logistic Regression were quite competitive as well, which implies that more simplified and understandable models can also be used in the context of clinical screening. The interpretable models often preferred decision in healthcare settings is more desirable since the clinicians can easily understand and depend on the process of decision-making. However, the close-to-perfect performance measures that are present in this study can as well reflect a possible limitation in the size of datasets, feature correlation, or

overfitting. The models recommended should therefore be verified in future researches with the assistance of larger and more diverse datasets which could be collected in multiple healthcare institutions.

In addition, longitudinal information about maternal health, behavioral measures, and electronic health records may also be integrated and potentially increase the power of prediction and be able to detect risk of postpartum depression earlier.

All in all, this study confirms that the decision-support systems that are trained on the information of questionnaires and run on the machine learning models are effective in detecting postpartum depression early enough and, as such, can intervene to guarantee improved maternal mental health.

VI CONCLUSION

The study is based on a specific dataset on maternity health that was gathered by use of questionnaires, and the group of respondents was mothers that are in the postpartum stage in Bangladesh. The data set contains information regarding the different demographics, socio-economic, obstetric, psychological, and the psycho-social factors related to perinatal depression. This data contains several variables, some of which are numerical and others are categorical which are all linked to one particular case, in this case, a woman in the post-partum condition. The Edinburgh postnatal depression scale (EPDS) was applied to determine the cases of the postpartum depression. In line with medical requirements regarding the issue, an end score of more than 13 on the EPDS was a pointer of having postpartum depression. To serve the analysis, a binary outcome variable was introduced (named as *ppd*) with a score of one (1) to indicate the presence and zero (0) to indicate the absence of postpartum depression. The data set had numerical variables like age, number of pregnancies, PHQ-9 score and EPDS score among other variables that described the type of relationship between the mother and infant

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